



CJC
Complete Java Classes

ABSTRACT CLASS (ABSTRACTION) TASK: -

List of tasks related to Abstract Class (Abstraction)–

Question 1. Define abstract class. Write the rules of abstract class?

Question 2. Write the difference between normal class and abstract class?

Question 3. Write the Difference between abstract class and interface?

Question 4. How do you declare an abstract class in Java?

Question 5. Check the program is it correct or not, if it is correct then write the output of the program.

```
public abstract class A {
```

```
    public void m1() {
```



CJC
Complete Java Classes

```
        System.out.println("m1_____A");
    }

}

public class B extends A {
    public void m2() {
        System.out.println("m2_____B");
    }
}

public class Test {
    public static void main (String args[]) {
        B b = new B ();
        b.m1();
        b.m2();
        A a = new B ();
        a.m1();
    }
}
```



Question 6. Complete the program, Call the methods and write the output of the program.

```
public abstract class A {
```

```
    abstract void m1();
```

```
}
```

```
public class B extends A {
```

```
}
```

```
public class C extends A {
```

```
}
```

```
public class Test {
```

```
    public static void main (String args[]){
```

```
        A a=new C ();
```

```
        A a1=new B ();
```

```
    }
```

```
}
```

Question 7. Check the program correct or not. If is it not then Why?



CJC
Complete Java Classes

```
public abstract class Parent {  
    public abstract final void m1();  
    public void m2 () {  
    }
```

```
}
```

```
public class child extends parent {  
    public final void m1() {  
    }  
}
```

```
public class Test {  
    public static void main (String args[]) {  
        Child c = new Child ();  
        c.m1();  
        c.m2();  
    }  
}
```



Question 8. Check the program correct or not. If is it not then Why?

```
public interface Mul {
```

```
    public abstract int mul1(int a, int b);
```

```
    public int mul2(int a, int b, int c);
```

```
    public void display () {
```

```
        System.out.println("Dispaly____method");
```

```
    }
```

```
}
```

```
public class Test implements Mul {
```

```
    public int mul1(int i1, int i2) {
```

```
        return i1 * i2;
```

```
    }
```

```
    public int mul2(int i1, int i2, int i3) {
```

```
        return i1 * i2 * i3;
```

```
    }
```

```
    public static void main (String args[ ]) {
```

```
        Mul obj = new Test ();
```



CJC
Complete Java Classes

```
        System.out.println(obj.mul1(3, 7));  
        System.out.println(obj.mul2(1, 9, 1));  
    }  
}
```

Question 9. Complete the program. What will be the output?

```
public interface Calculator {
```

```
    public void add();
```

```
    public void sub();
```

```
    public void mul();
```

```
    public void div();
```

```
}
```

```
public class A implements Calculator{
```

```
    public void add(){
```

```
    }
```

```
}
```

```
public class B {
```

```
    public void sub(){
```



CJC
Complete Java Classes

```
    }  
    public void mul(){  
    }  
}  
  
public class C {  
    public void div(){  
    }  
}  
  
public class Test{  
    public static void main(String args[]){  
        //call all methods Here  
    }  
}
```

Question 10. Complete the program. What will be the output?

```
public class Calculator {  
    public abstract void add();  
    public abstract void sub();  
}
```



CJC
Complete Java Classes

```
public abstract void mul();  
public abstract void div();  
}  
public class Addition extends Calculator {  
    public void add(){  
    }  
}  
public class Subtraction {  
    public void sub(){  
    }  
public class Multiplication {  
    public void mul(){  
    }  
}  
public class Division {  
    public void div(){  
    }  
}
```




CJC
Complete Java Classes

```
public class Test {
```

```
    public static void main(String args[]){
```

```
        //Call all methods Here
```

```
    }
```

```
}
```

THANK YOU



CJC
Complete Java Classes